DSA210 – Introduction to Data Science

Final Report

Exploring the Impact of Coffee Consumption on Sleep and Stress

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1. Motivation

Coffee is one of the most widely consumed beverages worldwide, yet its effects on sleep quality and stress levels are often discussed anecdotally rather than examined with data. This project was motivated by a practical question: does drinking more coffee actually affect how long you sleep, and does more screen time before bed increase stress?

From a data science perspective, this topic offers the opportunity to apply the full analysis pipeline — data collection, exploratory analysis, hypothesis testing, and machine learning — to a real and relatable problem. The self-collected nature of the dataset adds relevance: the data reflects actual behaviors of real participants, not an abstract benchmark.

2. Data Source

2.1 Collection Method

The dataset was collected through a structured Google Forms survey distributed to 106 participants in April 2026. All survey questions were mandatory. Participants reported their daily coffee habits, sleep patterns, stress levels, screen time before sleep, and exercise habits.

Attribute	Details
Participants	106 individuals
Method	Google Forms (self-reported)
Collection Period	April 2026
Target Group	Primarily university students and young adults
Significance level	α = 0.05

2.2 Variables

Variable	Type	Description
coffee	Numerical	Cups of coffee per day (0–5)
coffee_type	Categorical	Type of coffee preferred
sleep_hours	Numerical	Hours of sleep per night
sleep_quality	Ordinal (1–5)	Self-reported sleep quality
stress	Ordinal (1–5)	Self-reported stress level
screen_time	Numerical	Screen time before sleep (hours)
exercise	Categorical	Exercises regularly? (Yes/No)
age	Numerical	Age of participant
gender	Categorical	Gender of participant

2.3 Data Quality

Because all survey questions were mandatory, respondents who drink 0 cups of coffee per day were still required to select a coffee type. This resulted in inconsistent entries (e.g., coffee = 0 but coffee_type = 'Latte'). In preprocessing, the coffee_type of all non-drinkers (coffee = 0, n = 11) was corrected to 'Non-drinker'. No other missing values or duplicate rows were found in the dataset.

3. Data Analysis

3.1 Exploratory Data Analysis

The EDA phase covered distribution analysis for all numerical and categorical variables, a Pearson correlation matrix, box plots for spread and outlier detection, and bivariate visualizations between key variables.

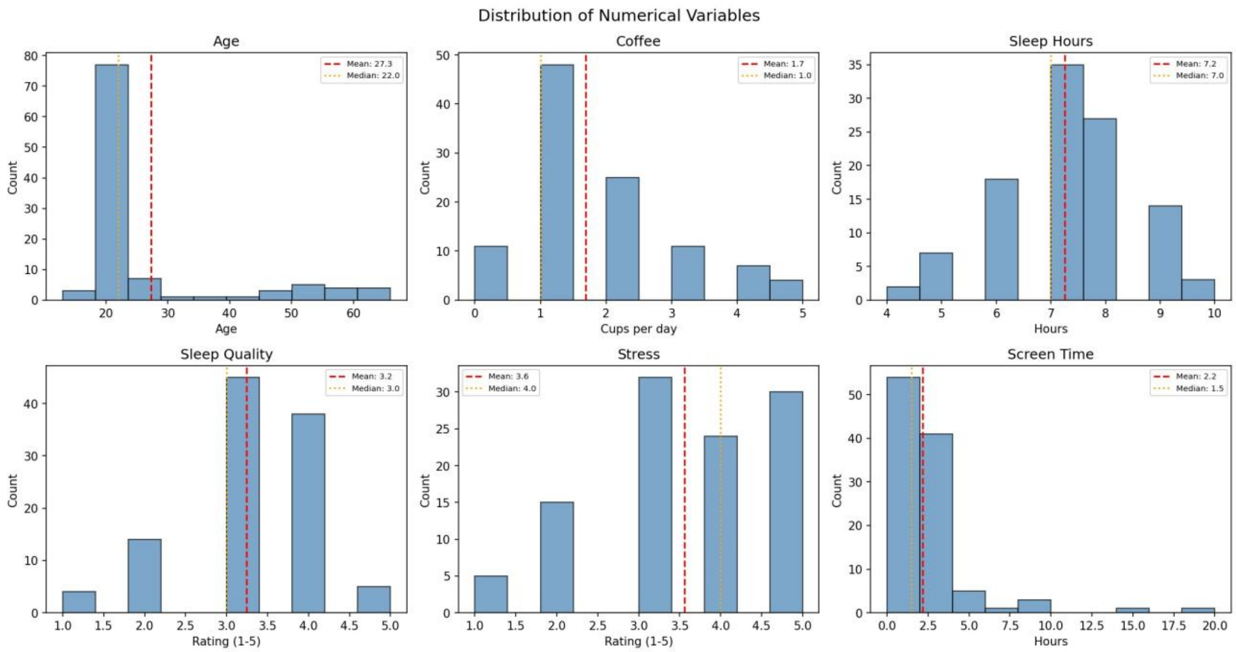

Figure 1: Distribution of numerical variables. Age and coffee are right-skewed; sleep hours are approximately normally distributed. Screen time shows a clear outlier at 20 hours.

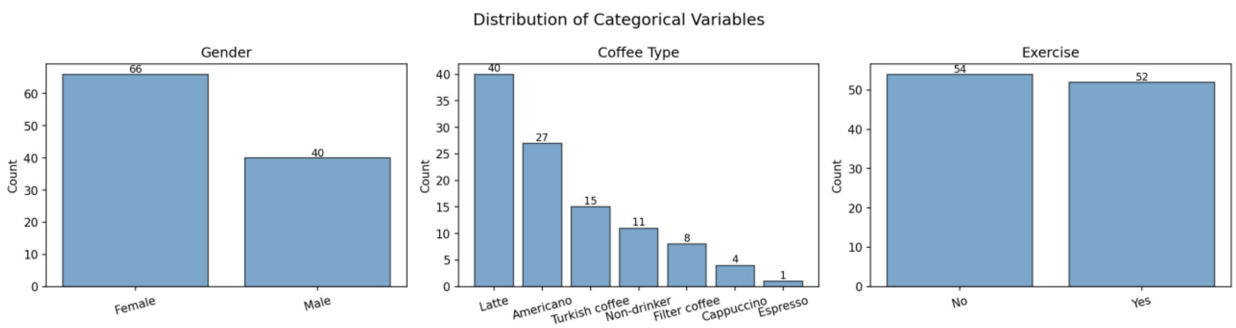

Figure 2: Distribution of categorical variables. The sample has more female participants (66 vs 40). Latte is the most common coffee type. Exercise habits are nearly evenly split.

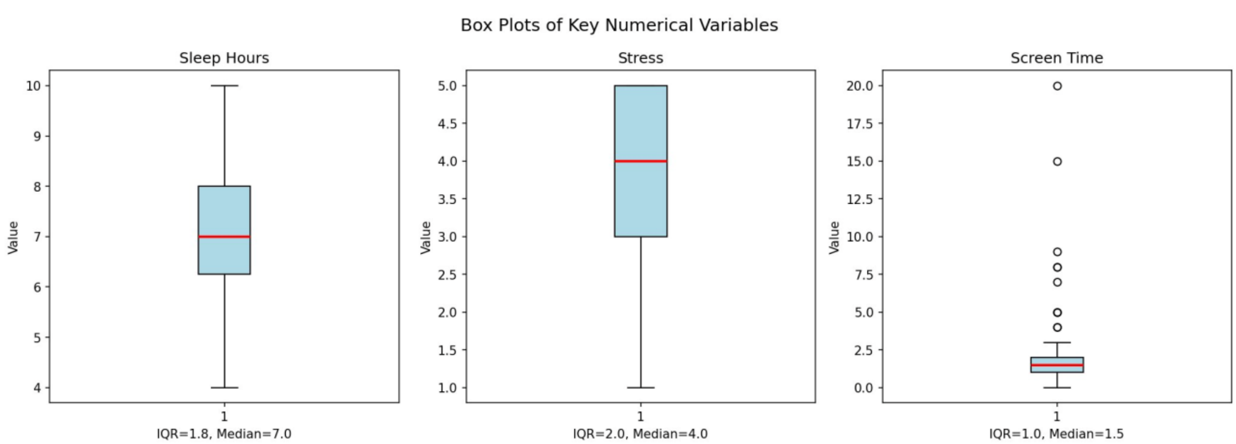

Figure 3: Box plots for sleep hours, stress, and screen time. The screen time outlier (20 hours) is clearly visible.

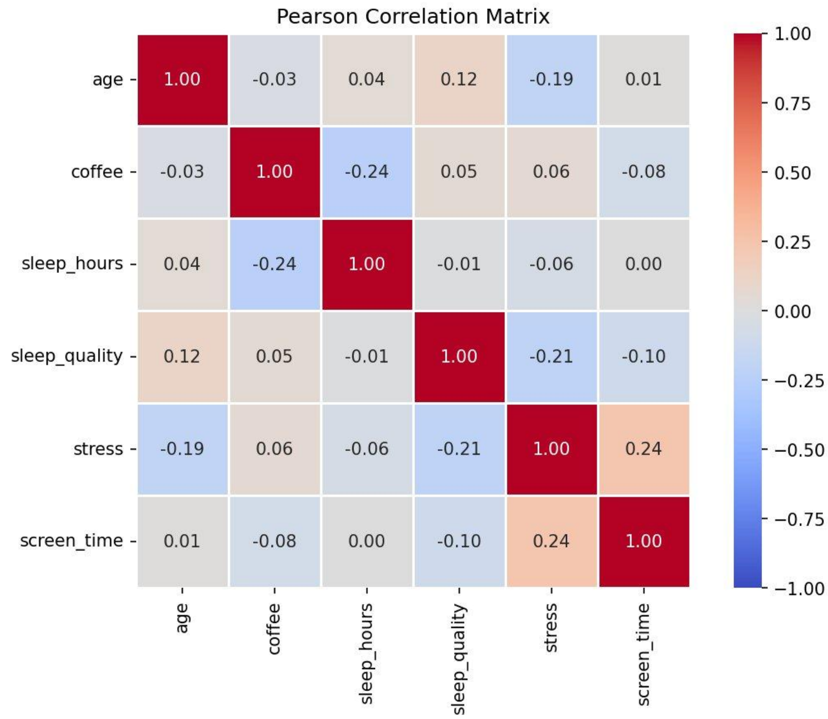

Figure 4: Pearson correlation matrix. The strongest correlations are coffee–sleep_hours (r = −0.24) and screen_time–stress (r = 0.24).

3.2 Hypothesis Testing

Four hypotheses were tested at a significance level of $\alpha = 0.05$:

#	Hypothesis	Test	Result	Finding
H1	Coffee → Sleep Hours	Pearson r	$p = 0.013$ ✓	$r \approx -0.24$, significant
H2	Coffee → Stress	Spearman ρ	$p > 0.05$ ✗	No significant relationship
H3	Exercise → Sleep Hours	Mann-Whitney U	$p > 0.05$ ✗	No significant difference
H4	Screen Time → Stress	Spearman ρ	$p < 0.05$ ✓	$\rho > 0$, significant

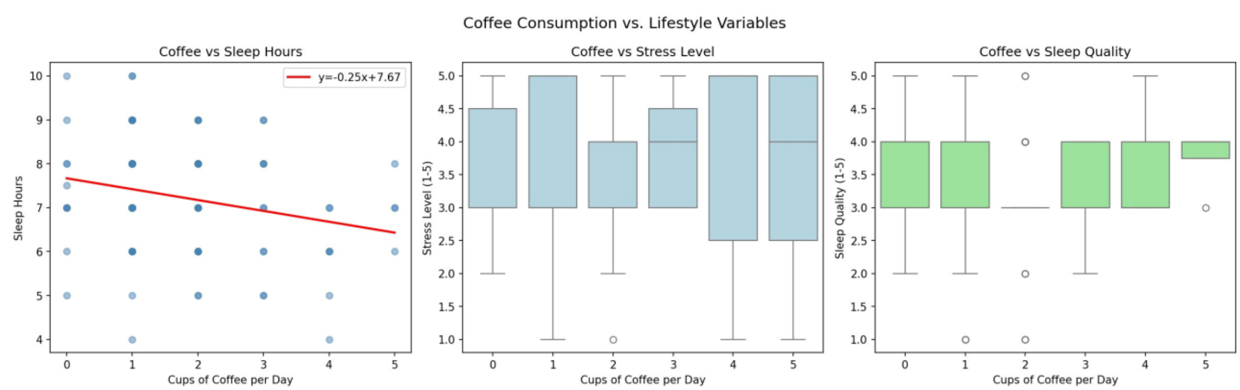

Figure 5: Coffee vs lifestyle variables. Left: negative slope confirms H1 (more coffee → less sleep). Middle and right: no clear pattern for stress and sleep quality, consistent with H2 not being significant.

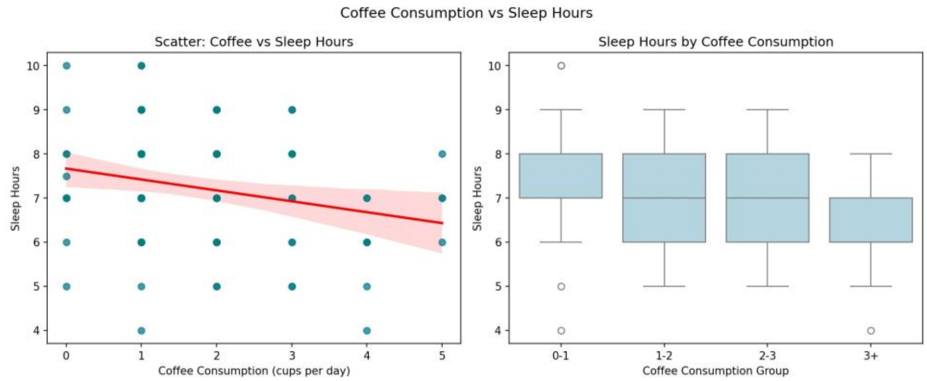

Figure 6: H1 — Coffee vs sleep hours with regression line and grouped box plots. The downward trend is consistent across all consumption groups.

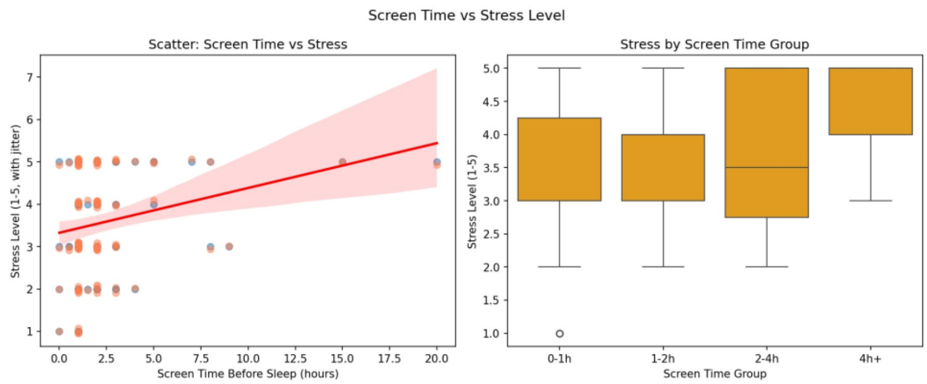

Figure 7: H4 — Screen time vs stress. The scatter plot shows a positive trend; the box plot confirms stress increases with screen time group.

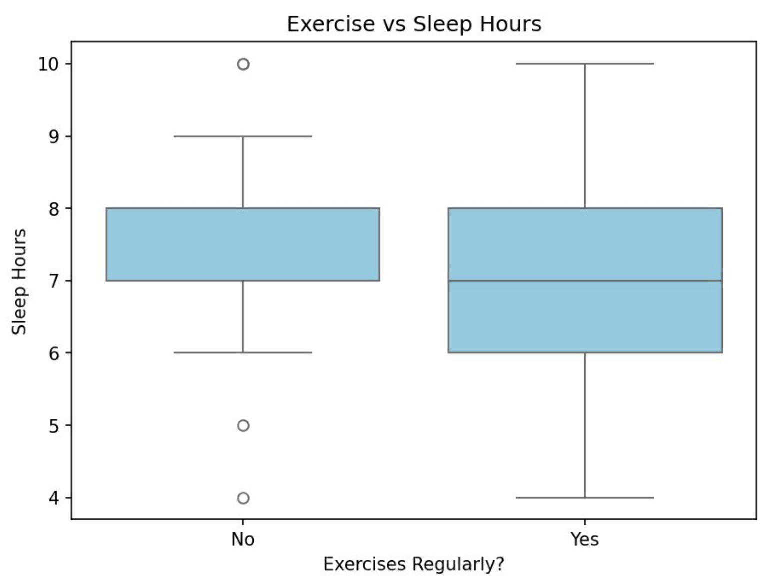

Figure 8: Exercise vs sleep hours (H3 — Mann-Whitney U, $p > 0.05$). No significant difference in sleep duration between exercisers and non-exercisers.

4. Machine Learning

4.1 Approach

The two statistically significant findings (H1 and H4) were formalized as supervised regression tasks. **Linear Regression** was selected as the primary model due to interpretability and suitability for small datasets. A **Decision Tree Regressor (max depth = 3)** was included as a comparison. An 80/20 train/test split and StandardScaler were applied consistently across both models. 5-fold cross-validation was used to assess generalization stability.

Model	Target	Features	EDA link
Model 1	sleep_hours	coffee, stress, screen_time, age	H1
Model 2	stress	screen_time, coffee, age	H4

4.2 Model 1: Predicting Sleep Hours

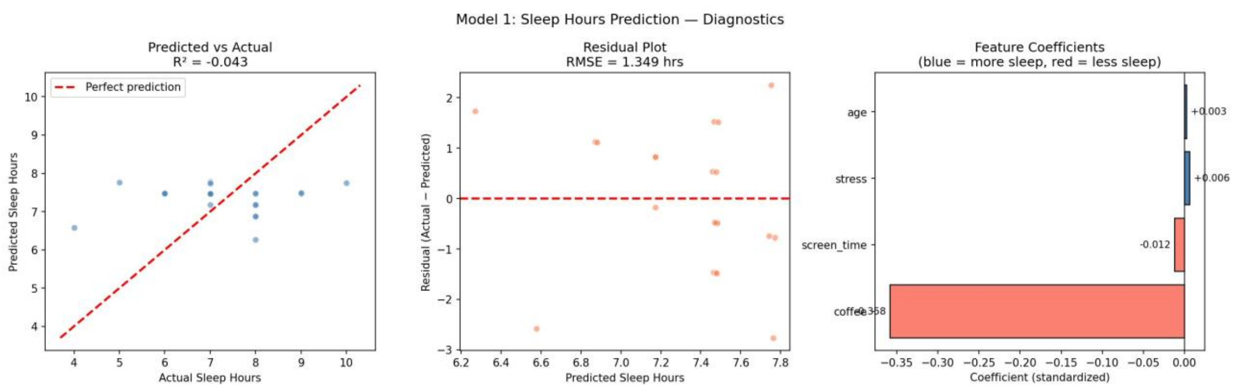

Figure 9: Model 1 diagnostics. Left: predicted vs actual (wide scatter, $R^2 = -0.043$). Middle: residuals show no strong pattern. Right: coffee has the largest negative coefficient, consistent with H1.

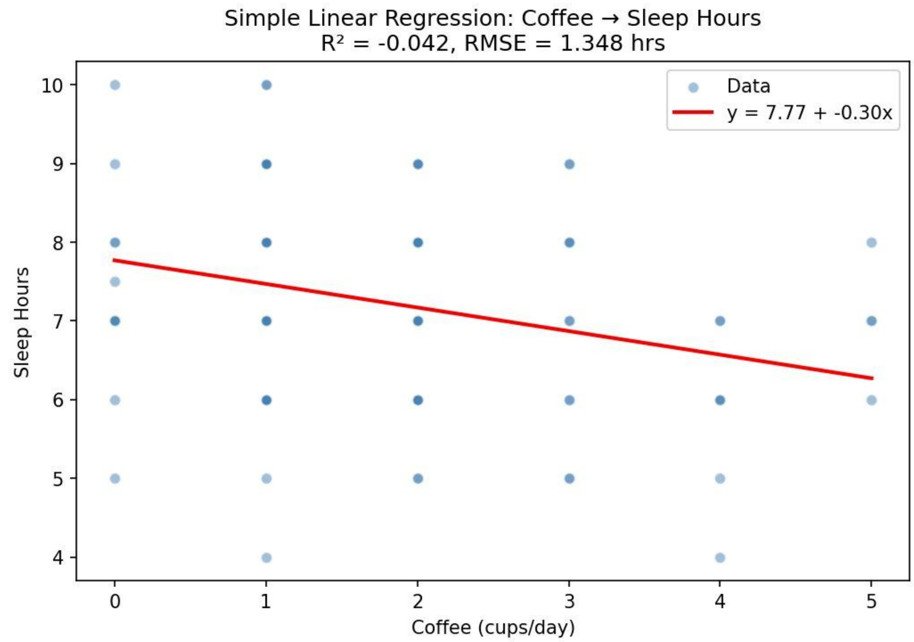

Figure 10: Simple linear regression using only coffee as a predictor. y = 7.77 − 0.30x: each additional cup is associated with approximately 0.30 fewer hours of sleep.

4.3 Model 2: Predicting Stress Level

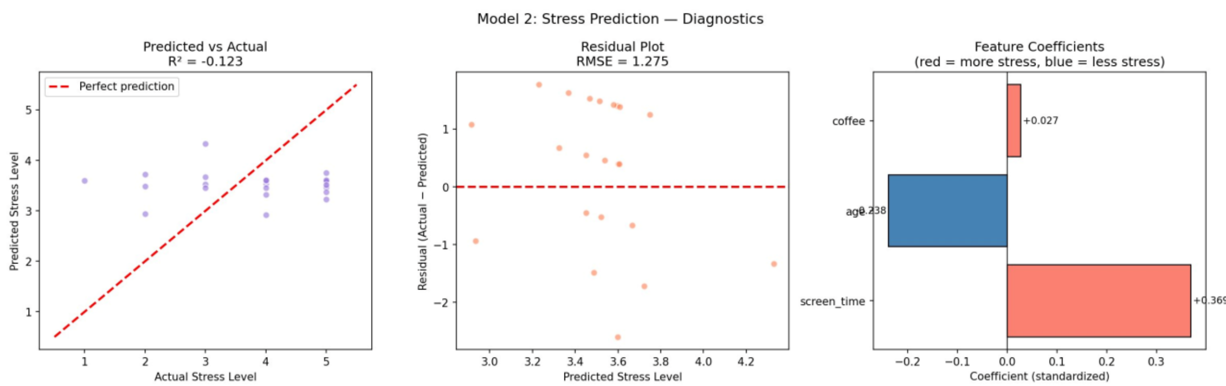

Figure 11: Model 2 diagnostics. screen_time has the largest positive coefficient (β = +0.369), consistent with H4. R^2 = −0.123.

4.4 Decision Tree Comparison

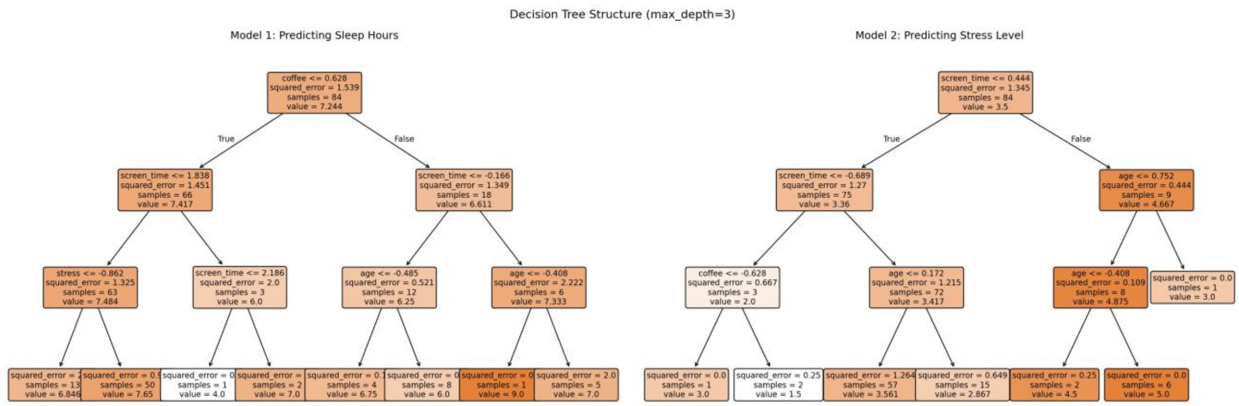

Figure 12: Decision Tree structure (max depth = 3). Model 1 first splits on coffee ≤ 0.628; Model 2 first splits on screen_time ≤ 0.444. Both confirm coffee and screen_time as the most informative features.

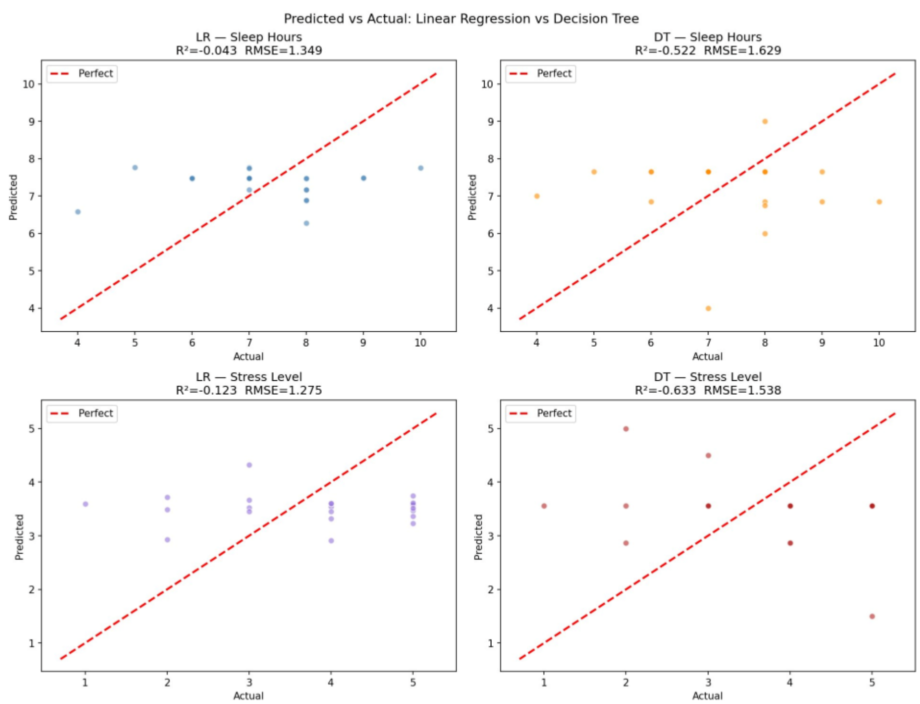

Figure 13: Predicted vs actual for all four model–target combinations. Points cluster around the mean in all cases, confirming that low predictive power is consistent across model types.

4.5 Interpretation

Why is R^2 low? Both Linear Regression and Decision Tree return near-zero or negative R^2 scores, meaning neither model predicts the target better than simply guessing the mean. This is expected: sleep duration and stress are shaped by many unmeasured factors (genetics, work schedule, mental health). A statistically significant correlation does not guarantee a high R^2 — they measure different things. With n = 106 self-reported responses, this result is honest and expected.

Directional consistency: Despite low R^2, coefficient signs are consistent with EDA findings. Coffee is negative in Model 1 (→ more coffee, less sleep) and screen_time is positive in Model 2 (→ more screen time, more stress). The Decision Tree's first splits also confirm these as the most informative features, further supporting the EDA conclusions.

Note on overfitting: The Decision Tree model is also susceptible to overfitting on small datasets. With only 84 training samples, a tree with max depth = 3 may capture noise rather than generalizable patterns. This further supports the use of Linear Regression as the primary model for this dataset.

5. Key Findings

- Coffee consumption is negatively associated with sleep duration ($r \approx -0.24$, $p = 0.013$). Statistically significant and consistent across EDA, hypothesis testing, and ML results.
- Screen time before sleep is positively associated with stress level (Spearman $\rho > 0$, $p < 0.05$), visible in grouped box plots and confirmed by the positive ML coefficient ($\beta = +0.369$).
- No statistically significant relationship was found between coffee and stress (H2), or between exercise and sleep hours (H3).
- Machine learning models confirm the direction of EDA findings but achieve low predictive power, expected for a small self-reported survey dataset.
- Decision Tree and Linear Regression perform similarly, confirming that the low predictive power is a data limitation, not a model problem.

6. Limitations and Future Work

6.1 Limitations

- Small sample size (n = 106) limits generalizability and machine learning model stability.
- Self-reported data is subject to recall bias and social desirability effects.
- Survey design forced responses on all questions, leading to inconsistencies corrected in preprocessing (n = 11 non-drinkers affected).
- The sample is predominantly young university students, limiting generalizability to other age groups.
- Correlation does not imply causation — observed associations may be driven by unmeasured confounders.

6.2 Future Work

- Collect more data (n > 500) for stable machine learning models with reliable cross-validation.
- Add features: caffeine timing, sleep environment, medication use, work and study hours.
- Use objective measurements (actigraphy, wearables) instead of self-reported data.
- Explore ensemble models (Random Forest, XGBoost) and kNN with a larger dataset.
- Apply time-series analysis if longitudinal data becomes available.

7. Conclusion

This project investigated the relationship between coffee consumption, sleep duration, stress level, and lifestyle habits using a self-collected survey dataset of 106 participants.

The analysis showed that higher coffee consumption is significantly associated with shorter sleep duration ($r \approx -0.24$, $p = 0.013$), while increased screen time before sleep is associated with higher stress levels. These findings were consistent across exploratory analysis, hypothesis testing, and machine learning.

Although machine learning models achieved low predictive performance ($R^2 < 0$), their feature coefficients and Decision Tree split structures remained directionally consistent with the statistical findings. The low R^2 reflects the inherent difficulty of predicting sleep and stress from a small self-reported survey — not a flaw in the methodology.

Overall, this project demonstrates how data science methods can be applied to a real-world behavioral dataset. It also highlights an important lesson: statistical significance and predictive power are distinct, and both must be interpreted carefully.

8. Ethical Considerations and AI Usage

8.1 Data Ethics

Survey responses were collected voluntarily through Google Forms. Although email collection was enabled during the response phase to prevent duplicate submissions, all email addresses and personally identifiable information were removed before preprocessing and analysis. The final dataset used in this project was fully anonymized.

8.2 AI Tool Usage

In accordance with the course guidelines, AI tools were used as supportive assistants during the project. Claude AI and ChatGPT were occasionally used for debugging Python code, improving markdown formatting, refining report language, and receiving feedback on data analysis workflows.

All preprocessing steps, hypothesis formulations, model selections, interpretations, and conclusions were independently designed, implemented, and verified by the author. AI tools were used only to support productivity and editing, not to replace analytical understanding or decision-making.